

Modern IOT Pager System Using ESP32 and Telegram Bot API

1. Abstract

The Modern IoT Pager System is a secure communication device designed for places where quick and reliable messaging is important, such as hospitals and industrial environments. It is built using the ESP32 microcontroller and connects to the Telegram Bot API through encrypted HTTPS using a sequential polling method. Instead of relying on traditional radio-based pager systems, this project uses a flexible Wi-Fi network to enable two-way communication. The system operates through a simple loop that manages text scrolling on a 16×2 LCD display, provides sound alerts based on the system state, and allows users to respond using manual buttons. By combining the reliability of traditional pagers with the flexibility of modern IoT technology, this solution offers an affordable and distraction-free alternative to smartphones while still ensuring secure and reliable data communication.

2. Introduction

In critical professional environments such as hospitals and secure industrial facilities, reliable and distraction-free communication is essential. Standard cellular networks often introduce notification fatigue and security risks, making traditional paging systems a necessity for high-priority alerts. However, legacy radio infrastructure lacks the flexibility of modern internet protocols and encrypted feedback.

This project proposes a low-cost, automated communication system using an **ESP32 microcontroller** and Telegram Bot API to facilitate secure two-way messaging. The system utilizes a sequential polling architecture to monitor incoming text, calculate message length for scrolling, and provide auditory alerts. If a message is received, the system triggers a visual display and allows the user to send predefined responses to ensure road-tested reliability in professional settings.

3. Problem Statement

The community faces the challenge of maintaining secure, real-time communication in sensitive zones where smartphones are restricted or unreliable.

Traditional paging methods are often inefficient, costly, and require specialized radio towers. There is a need for an automated, cost-effective, and reliable system to continuously monitor digital messages and issue auditory alerts in case of emergency communications.

4. Objectives of the Project

- To design and implement a secure IoT-based communication node using the ESP32 microcontroller and Telegram API.
- To integrate the ESP32 with the Telegram messaging platform for receiving messages in real time.
- To notify users of incoming messages through a buzzer alert and LCD display.
- To implement a scrolling text feature for displaying longer messages.
- To allow users to send predefined responses such as OK or NO using hardware buttons.

5. Literature Review

[1] Early pager technology was pioneered by Motorola, which introduced portable radio receivers capable of receiving broadcast alerts from centralized transmitters. These early devices primarily produced audible beeps to notify users of incoming messages. [2] Research on wireless communication systems explains that paging networks operate using dedicated radio frequencies that transmit signals from a base station to multiple receivers simultaneously, ensuring reliable message delivery over large areas. [3] According to general technical documentation, pagers later evolved from simple beep-only devices to alphanumeric pagers capable of displaying short text messages on small LCD screens, improving communication efficiency. [4] Recent embedded systems research highlights the development of modern pager-like devices using microcontrollers such as ESP32. The LILYGO T-LoRa Pager demonstrates how compact IoT hardware can transmit and receive short messages using wireless communication modules. [5] Studies in healthcare communication systems report that pagers remain widely used in hospitals due to their reliability, low power consumption, and ability to function effectively even in environments where cellular network coverage is weak.

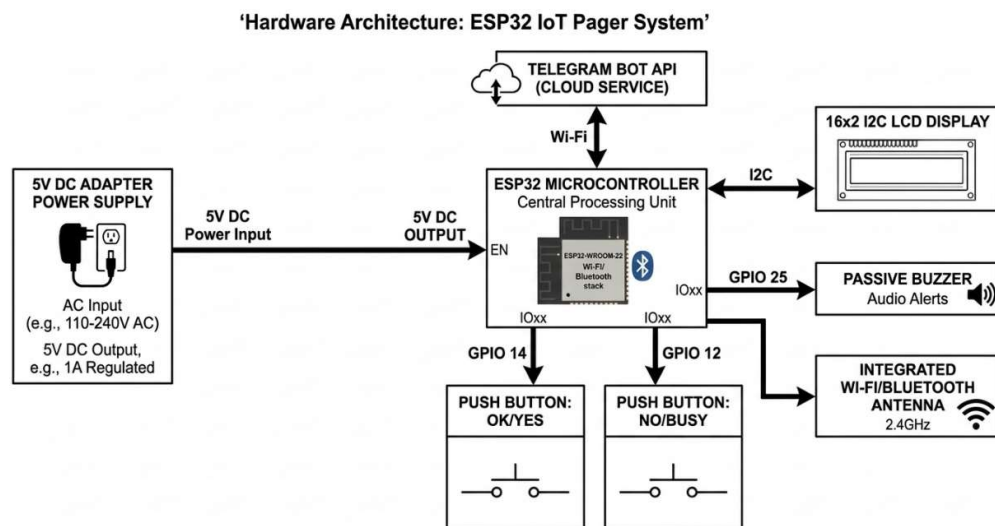
6. Proposed Methodology

The system uses the ESP32 microcontroller as the central processing unit. The ESP32 connects to a Wi-Fi network and communicates with the Telegram server using the Telegram Bot API.

The device continuously checks for new messages from the Telegram server using a polling mechanism. When a message is received, the system activates a buzzer notification and displays the message on a 16×2 LCD screen. If the message exceeds the display limit, a text rotation algorithm scrolls the message across the screen.

The system also includes two input buttons that allow the user to send predefined responses back to the Telegram chat. This enables two-way communication between the user and the device.

6.1 Block Diagram



6.2 Hardware Components

- 1) ESP32 microcontroller
- 2) Buzzer
- 3) LCD Display with I2C protocol
- 4) Wi-Fi Network Module (built into ESP32)
- 5) Power Supply Unit (5V Adapter)

6.3 Software Components

- 1) Arduino IDE: Used for programming the ESP32
- 2) UniversalTelegramBot Library: Facilitates the HTTPS communication with the Telegram API.
- 3) ArduinoJson Library: Used for parsing complex message data received from the cloud.

7. Expected Outcomes

The primary outcome of this project is a functional, handheld IoT pager that successfully establishes a secure Wi-Fi connection and monitors a specific Telegram bot for updates. The system will accurately display messages, with a rotating text function for strings exceeding 16 characters. Additionally, the system will demonstrate reliable two-way communication by allowing the user to transmit response packets back to the sender, confirmed by distinct audio melodies.

8. Applications

- Hospital emergency notification systems
- Industrial alert systems
- Office communication systems
- Smart home notification systems
- Security monitoring applications

References

- [1] [Motorola, History of Pager Technology.](#)
- [2] Institute of Electrical and Electronics Engineers, Wireless Communication Systems and Paging Networks.
- [3] “Pager,” Wikipedia, Online Encyclopedia.
- [4] [LILYGO T-LoRa Pager, ESP32-based pager communication device documentation.](#)
- [5] Ullah, E., Baig, M. M., GholamHosseini, H., & Lu, J., [Use of pager devices in public hospitals as a critical communication tool](#), Healthcare Communication Systems and Pager Reliability Studies.